BTEC Level 3 Applied Science · Unit 1 Biology · Answer sheet

Microscopy — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDScience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. M = image size / actual size. Resolution: smallest distance at which two points can still be seen as separate.

2. 2500 µm; 2.5×10^6 nm.

Equation	1 mm = 1000 µm = 1 000 000 nm
Substitution	2.5 mm × 1000
Working	2500 µm; 2.5 × 10⁶ nm
Answer + unit	2500 µm and 2.5 × 10⁶ nm

Marking guidance: 1 mark each conversion.

3. Eyepiece magnification × objective magnification.

4. 18 mm = 18 000 µm; actual = 18 000/600 = 30 µm.

Equation	A = I / M
Substitution	I = 18 mm = 18 000 µm; M = 600
Working	A = 18 000 / 600
Answer + unit	30 µm

Marking guidance: 1 mark conversion; 1 mark 30 µm.

5. 3.0 cm = 30 000 µm; M = 30 000/7.5 = ×4000.

Equation	M = I / A
Substitution	I = 3.0 cm = 30 000 µm; A = 7.5 µm
Working	M = 30 000 / 7.5
Answer + unit	×4000

Marking guidance: 1 mark conversion; 1 mark ×4000.

6. 40 mm = 40 000 µm; M = 40 000/10 = ×4000.

Equation	M = I / A
Substitution	I = 40 mm = 40 000 µm; A = 10 µm
Working	M = 40 000 / 10
Answer + unit	×4000

Marking guidance: 1 mark conversion; 1 mark ×4000.

7. 35 mm = 35 000 µm; actual = 70 µm. [2]

Equation	A = I / M
Substitution	I = 35 mm = 35 000 µm; M = 500
Working	A = 35 000 / 500
Answer + unit	70 µm

Marking guidance: 1 mark conversion; 1 mark 70 µm.

8. Light about 0.2 µm; TEM much smaller (about 0.1 nm) because electrons have a shorter wavelength. [4]

9. Living specimens / colour / cheaper / easier preparation. [2]

10. The granules are closer together than the resolution. Extra magnification enlarges the blur (empty magnification).